

```

top - 12:22:39 up 69 days, 10:28, 1 user, load average: 0.06, 0.13, 0.09
tasks: 187 total, 2 running, 305 sleeping, 0 stopped, 0 zombie
Cpu(s): 32.0%us, 1.7%sy, 0.0%ni, 71.7%id, 14.7%wa, 0.0%st, 0.0%rs
Mem: 1048576k total, 1025676k used, 22300k free, 14952k buffers
Swap: 4501528k total, 586552k used, 3984976k free, 150812k cached

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
15484	apache	15	0	359m	78m	35m	0	3.7	6.9	1:43.43	httpd
26843	apache	16	0	364m	75m	35m	0	3.7	7.4	0:59.08	httpd
15809	root	15	0	442m	75m	274m	5	1.1	0.9	30:11.17	mysqld
1	root	15	0	10364	384	332	0	0.0	0.0	0:10.02	init
2	root	RT	-5	0	0	0	0	0.0	0.0	0:00.00	migration/0
3	root	34	19	0	0	0	0	0.0	0.0	0:00.17	ksoftirqd/0
4	root	RT	-5	0	0	0	0	0.0	0.0	0:00.00	watchdog/0
5	root	10	-5	0	0	0	0	0.0	0.0	0:00.26	events/0
6	root	10	-5	0	0	0	0	0.0	0.0	0:00.00	khelper
7	root	11	-5	0	0	0	0	0.0	0.0	0:00.02	kthread
9	root	10	-5	0	0	0	0	0.0	0.0	0:00.00	xenwatch
10	root	10	-5	0	0	0	0	0.0	0.0	0:00.00	xenbus
15	root	10	-5	0	0	0	0	0.0	0.0	0:00.00	kblockd/0
16	root	20	-5	0	0	0	0	0.0	0.0	0:00.00	cqunev/0
20	root	20	-5	0	0	0	0	0.0	0.0	0:00.00	khubd
22	root	10	-5	0	0	0	0	0.0	0.0	0:00.00	kcrld
84	root	15	0	0	0	0	0	0.0	0.0	0:00.00	kbugtaskd
87	root	11	-5	0	0	0	0	0.0	0.0	0:22.97	kswapd0
88	root	20	-5	0	0	0	0	0.0	0.0	0:00.00	ain/0
218	root	11	-5	0	0	0	0	0.0	0.0	0:00.00	kpsmouse
219	root	10	-5	0	0	0	0	0.0	0.0	0:00.00	kstrpid
248	root	20	-5	0	0	0	0	0.0	0.0	0:00.00	kined
259	root	10	-5	0	0	0	0	0.0	0.0	1:02.04	kjournald
286	root	10	-5	0	0	0	0	0.0	0.0	0:00.60	kauditd
314	root	16	-4	13732	384	384	5	0.0	0.0	0:00.24	udsvie
640	root	12	-5	0	0	0	0	0.0	0.0	0:00.00	kpostd/0
650	root	12	-5	0	0	0	0	0.0	0.0	0:00.00	knapthandlerd
670	root	10	-5	0	0	0	0	0.0	0.0	0:00.00	kjournald
834	root	12	-5	0	0	0	0	0.0	0.0	0:00.00	iscst_sh
856	postfix	15	0	54252	2324	1880	5	0.0	0.2	0:00.00	trivial-rewrite

Figure 2: Memory

free up RAM for Apache. Figure 2 shows a screenshot from one of our servers.

Look for unnecessary OS processes that are running and consuming memory. From the listing, right at the bottom, you can see that 'postfix' is running. Postfix is a mail server that was installed as part of the distribution and is not required to be on the same server as the site. You can move services like this to a separate server and free up resources for the site.

Use the Drupal modules that inhibit performance, selectively

Use `ps -aux | sort -u` and `netstat -anp | sort -u` to check the processes and their resource consumption. If you find that the Apache process size is unduly high, the first step is to disable unnecessary modules on the production servers to free memory. Modules like Devel are required only in your development environment and can be removed altogether. After you have disabled unnecessary processes and removed unnecessary modules, run the `top` command again to check if the changes you have made have taken effect.

Reduce Apache's memory footprint

The Web server is installed as part of your distribution and can contain modules that you might not require, for example, Perl, etc. Disable modules that are not required. On Ubuntu, you can disable an Apache module using `a2dismod` while on Centos machines, you'll have to rename the appropriate configuration file in `/etc/httpd/conf.d/`. On Centos, some modules are loaded by directives in `/etc/httpd/conf/httpd.conf`. Comment out the modules (look for the `LoadModule` directive) that you

don't need. This will reduce the memory footprint considerably. On some of our installations, we disable status and mod-cgi because we don't need the server status page, and we run PHP as an Apache module and not as CGI.

Configure Apache for optimum performance

Another raging debate is on whether to use Worker or Prefork. Our Apache server uses the non-threaded MPM Prefork instead of the threaded Worker. Prefork is the default setting and we have left it at that. We haven't had issues with this default configuration and hence aren't motivated enough to try Worker. You can check what you are running using the following commands:

```

/usr/sbin/httpd -l (for Centos) and..
/usr/sbin/apachectl -l (for Ubuntu).

```

Set the `HostnamesLookups` to 'Off' to prevent Apache from performing a DNS lookup each time it receives a request. The downside is that the Apache log files will have the IP address instead of the domain name against each request.

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